

Announcement

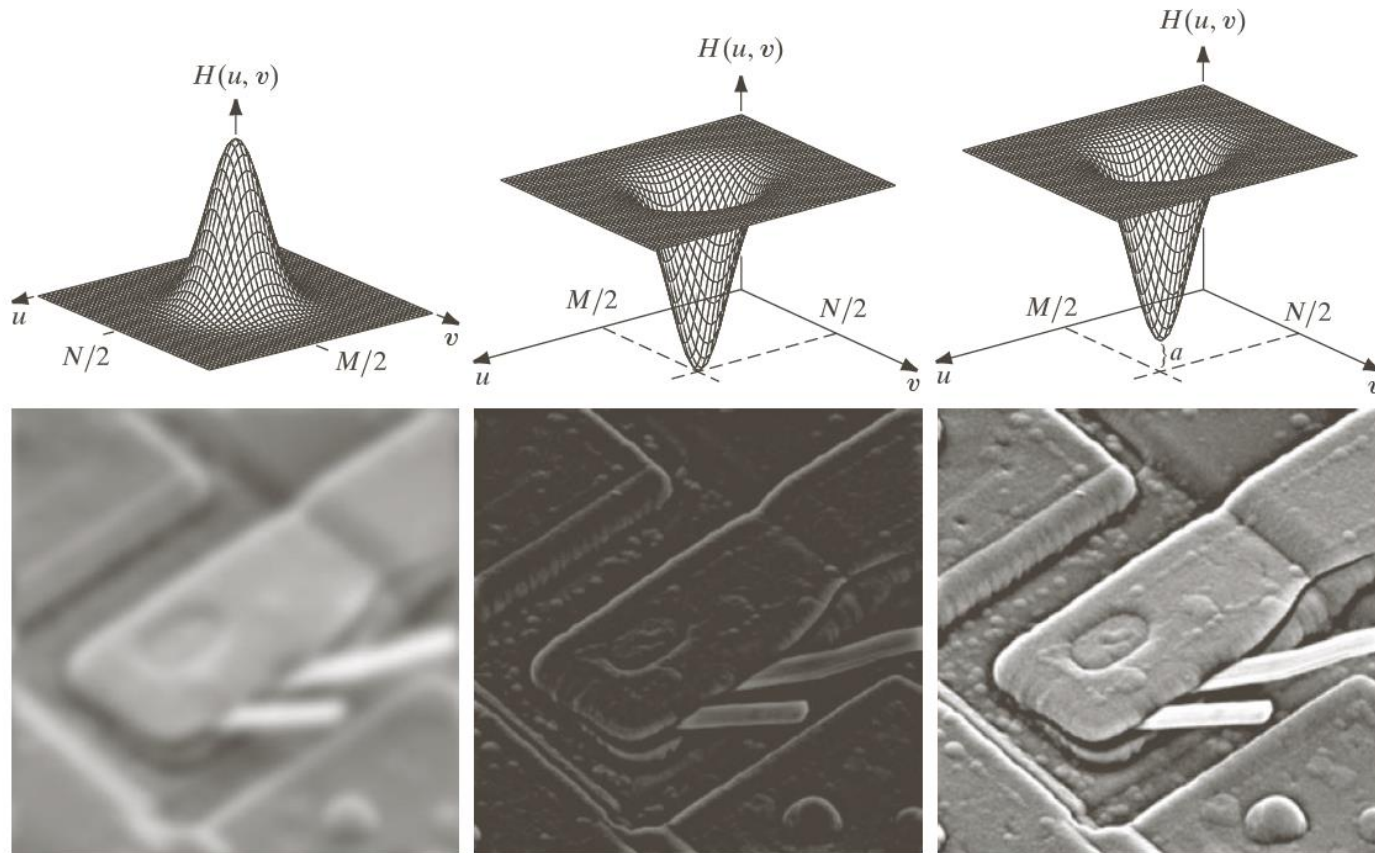
Homework #4 has been posted on Blackboard and course website.

Due 1:15pm, Thursday, Feb. 26th

Today's Agenda

- **Filtering in Frequency domain**

Low-Pass and High-Pass Filters



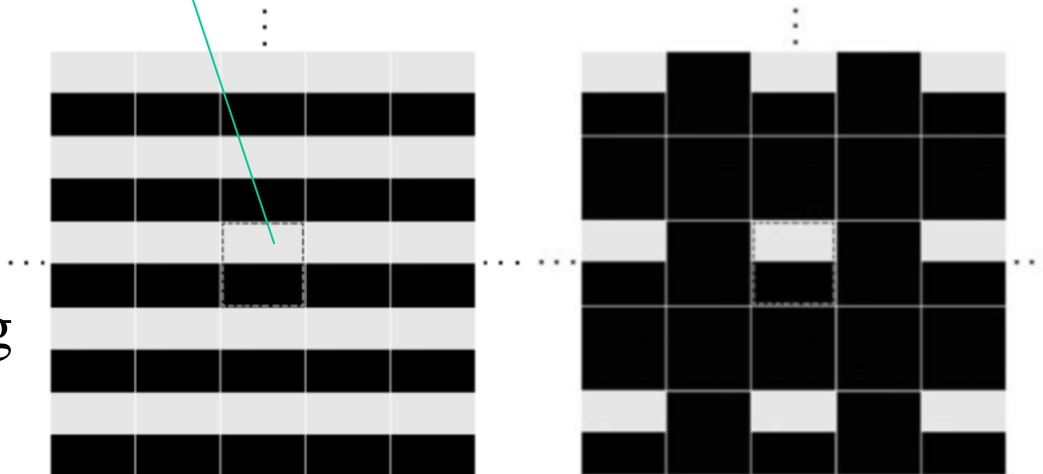
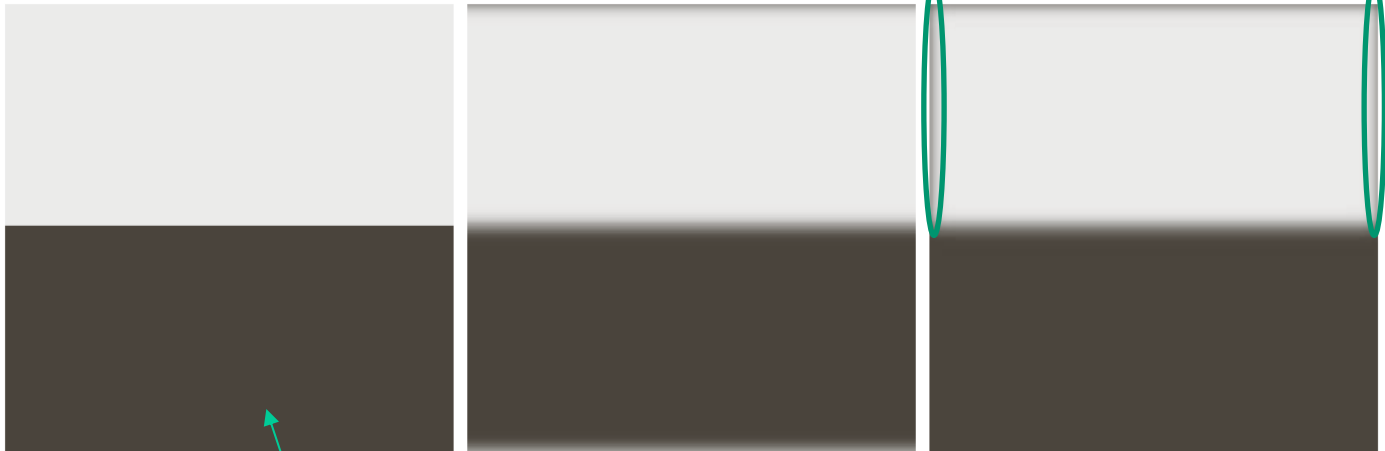
a	b	c
d	e	f

FIGURE 4.31 Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used $a = 0.85$ in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).

Effects of Zero-Padding

Smoothing, no padding

Smoothing, padded



No padding

Padded

Spatial Zero-Padding for the Filter

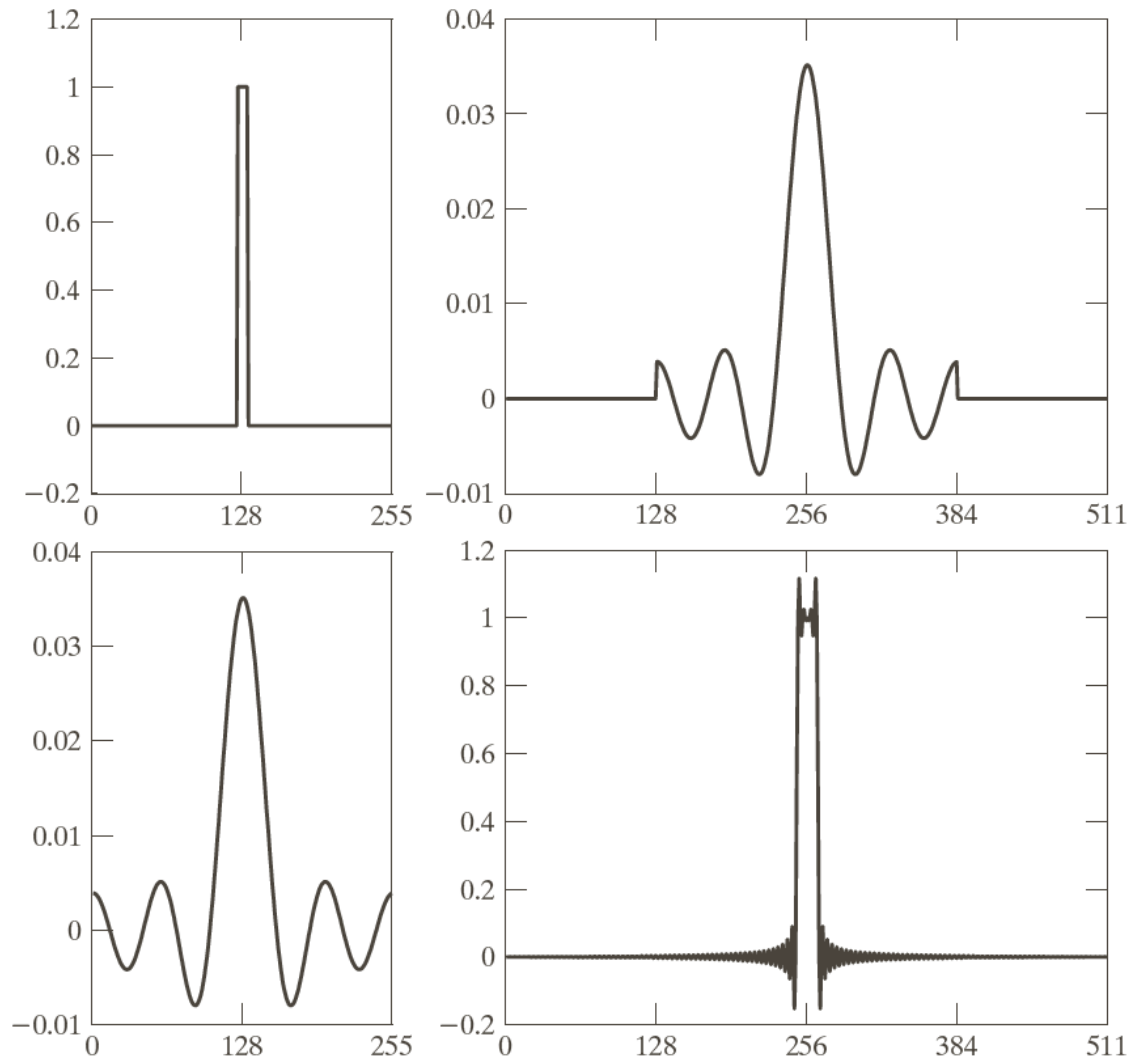


FIGURE 4.34

(a) Original filter specified in the (centered) frequency domain. (b) Spatial representation obtained by computing the IDFT of (a). (c) Result of padding (b) to twice its length (note the discontinuities). (d) Corresponding filter in the frequency domain obtained by computing the DFT of (c). Note the ringing caused by the discontinuities in (c). (The curves appear continuous because the points were joined to simplify visual analysis.)

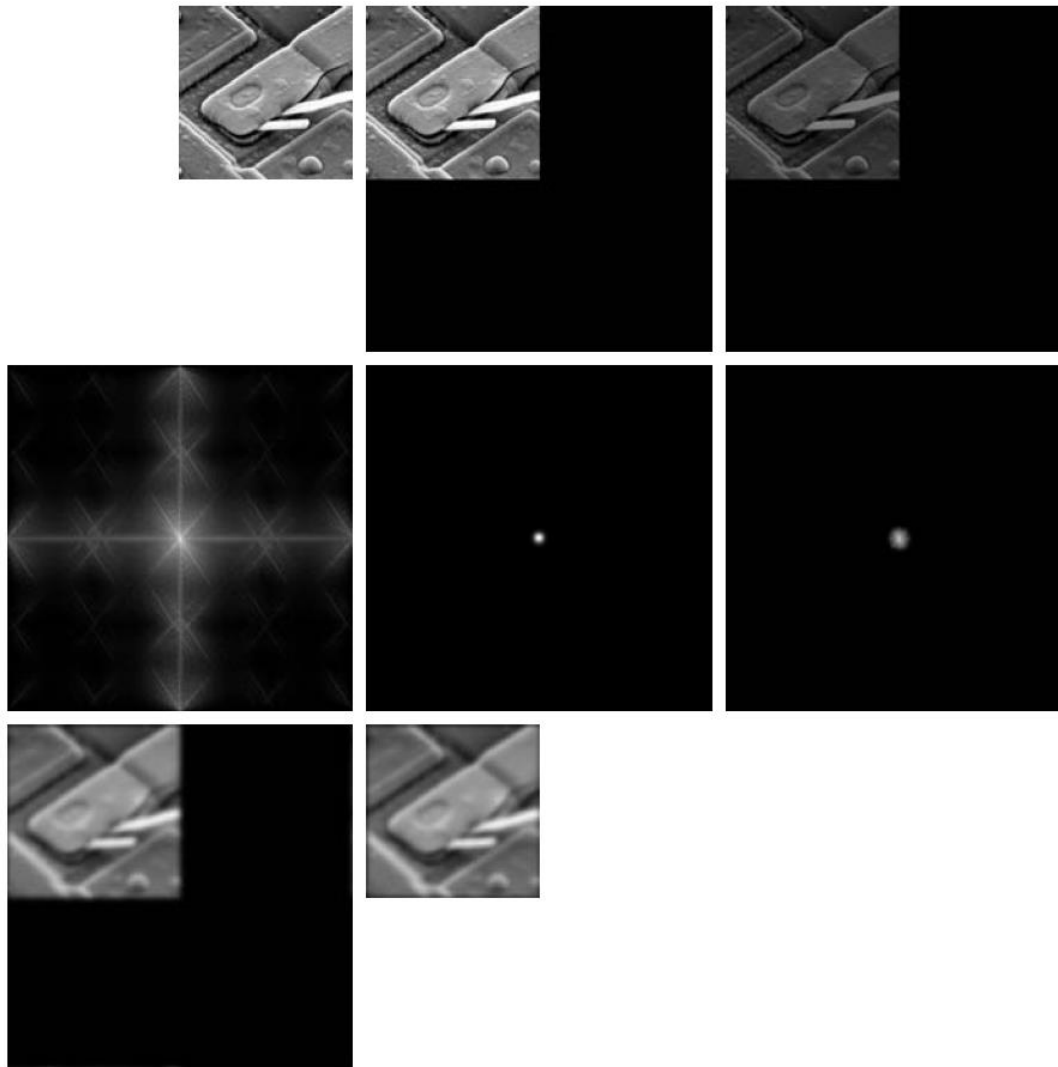
Summary: Filtering in the Frequency Domain

1. For $f(x, y)$, find $P = 2M, Q = 2N$
2. Form a padded image $f_p(x, y)$
3. Centering: Multiply $f_p(x, y)$ by $(-1)^{x+y}$
4. Compute DFT $F(u, v)$
5. Generate a real symmetric filter $H(u, v)$ of a size $P \times Q$, centered at $(P/2, Q/2)$ and make $G(u, v) = H(u, v)F(u, v)$
6. Obtain the processed image (should be real in theory)

$$g_p(x, y) = \{\text{real}[g(x, y)]\}(-1)^{x+y}$$

7. Obtained the final processed results $g(x, y)$ from the top left $M \times N$ region

An Example

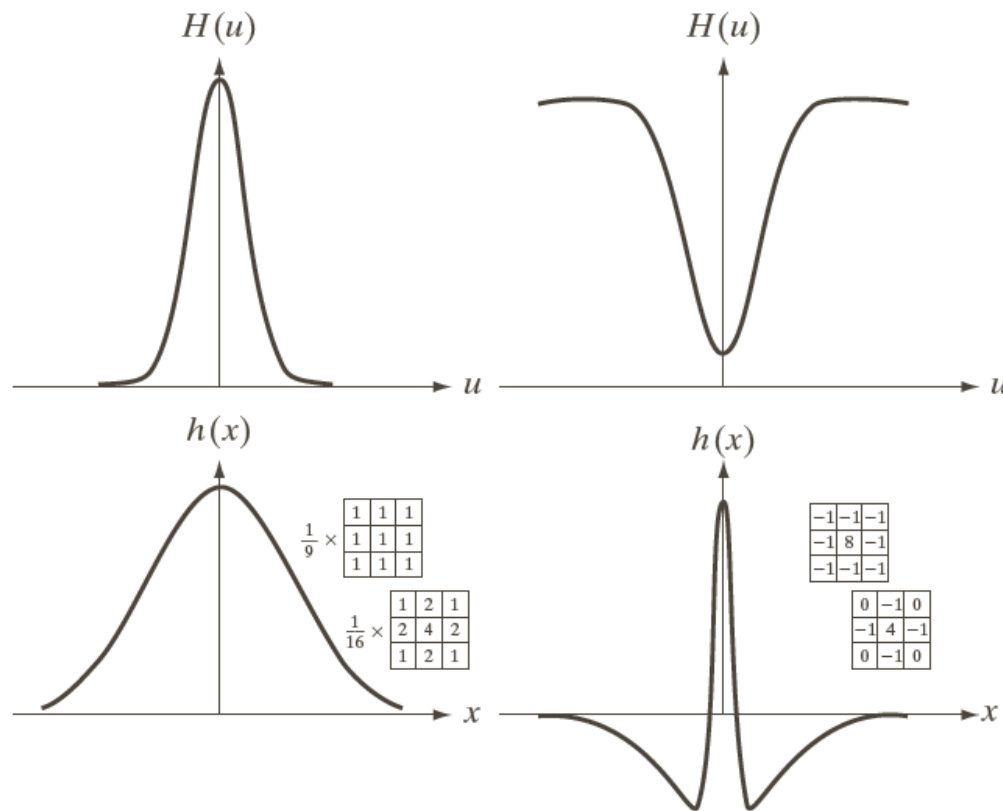


a	b	c
d	e	f
g	h	

FIGURE 4.36

- (a) An $M \times N$ image, f .
- (b) Padded image, f_p of size $P \times Q$.
- (c) Result of multiplying f_p by $(-1)^{x+y}$.
- (d) Spectrum of F_p .
- (e) Centered Gaussian lowpass filter, H , of size $P \times Q$.
- (f) Spectrum of the product HF_p .
- (g) g_p , the product of $(-1)^{x+y}$ and the real part of the IDFT of HF_p .
- (h) Final result, g , obtained by cropping the first M rows and N columns of g_p .

Correspondence to the Spatial Domain Filter



a	c
b	d

FIGURE 4.37

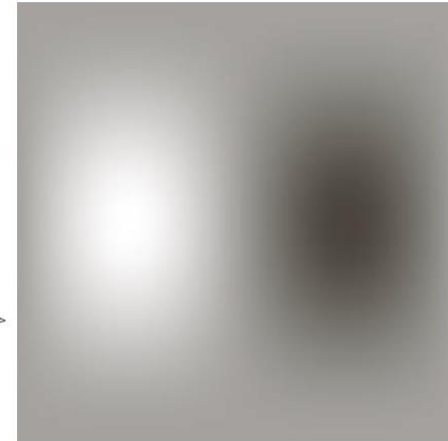
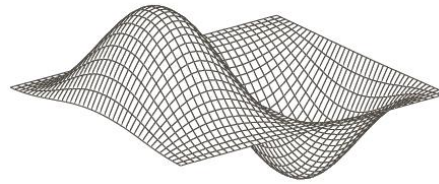
(a) A 1-D Gaussian lowpass filter in the frequency domain. (b) Spatial lowpass filter corresponding to (a). (c) Gaussian highpass filter in the frequency domain. (d) Spatial highpass filter corresponding to (c). The small 2-D masks shown are spatial filters we used in Chapter 3.

The FT of a Gaussian function is still a Gaussian function

An Example (Sobel Mask)



-1	0	1
-2	0	2
-1	0	1



a	b
c	d

FIGURE 4.39

(a) A spatial mask and perspective plot of its corresponding frequency domain filter. (b) Filter shown as an image. (c) Result of filtering Fig. 4.38(a) in the frequency domain with the filter in (b). (d) Result of filtering the same image with the spatial filter in (a). The results are identical.

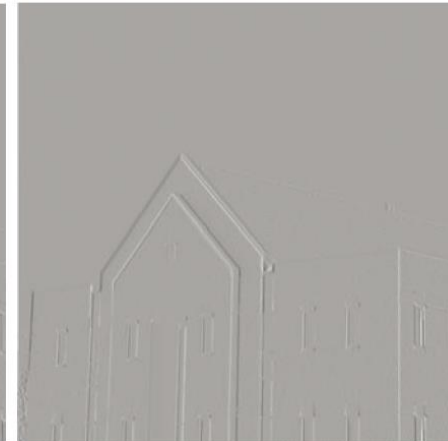
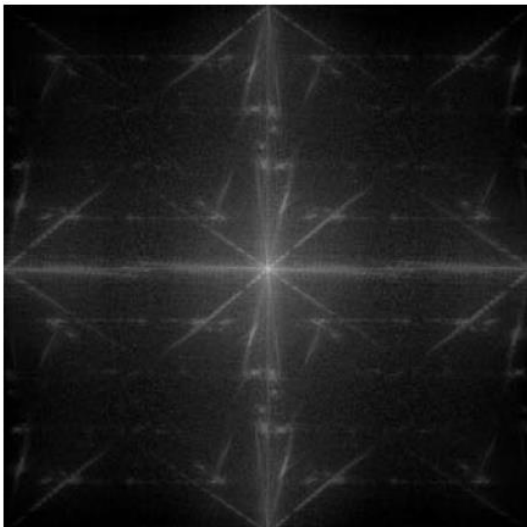
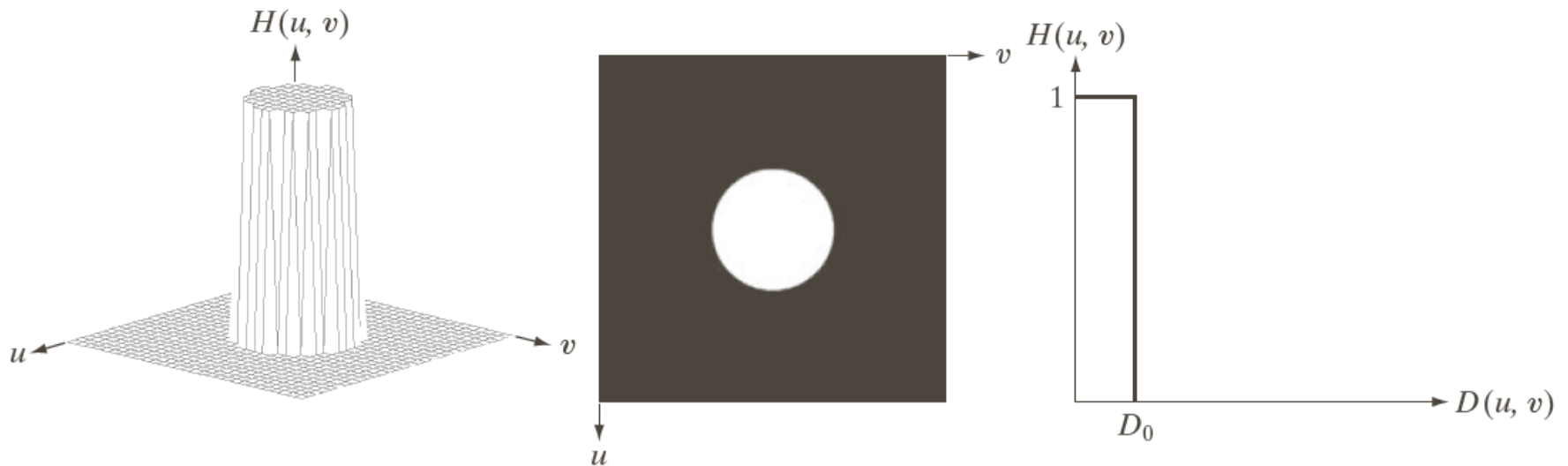


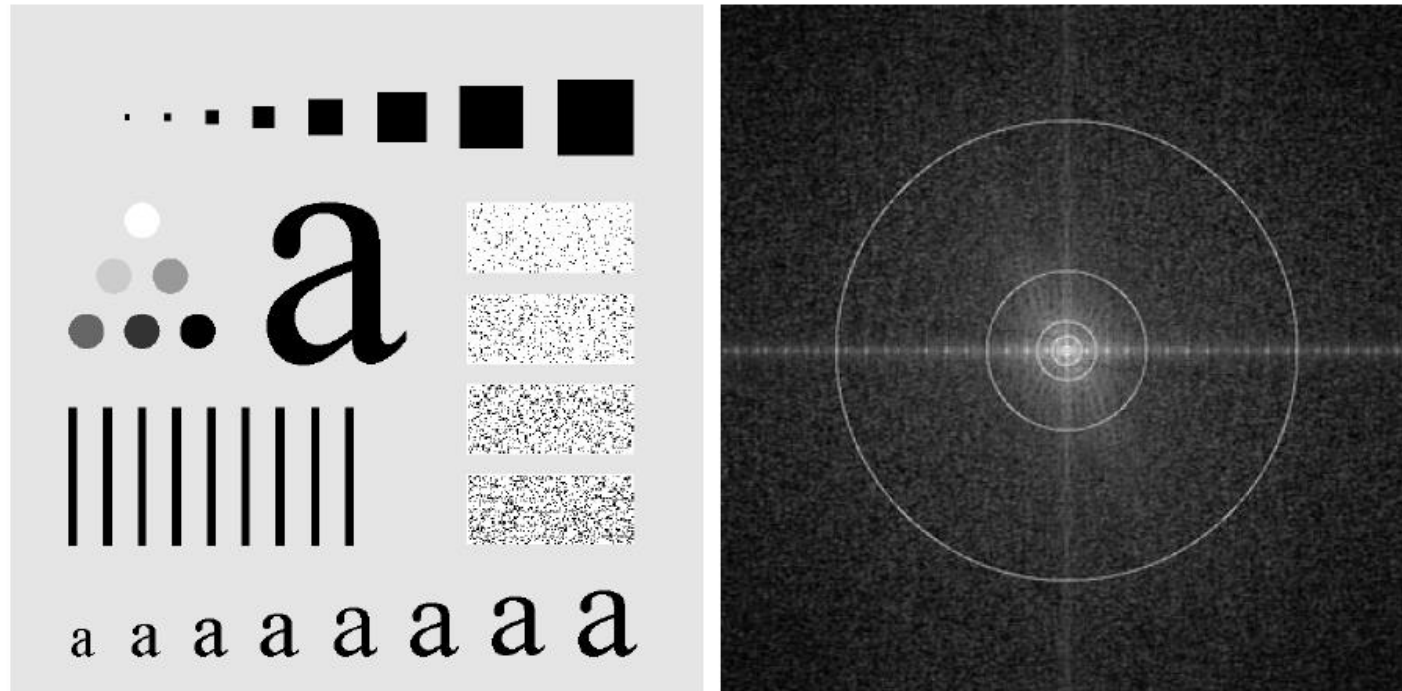
Image Smoothing Using Frequency Domain Filters – Ideal Lowpass Filter



a b c

FIGURE 4.40 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

Locating the Cut-Off Frequency

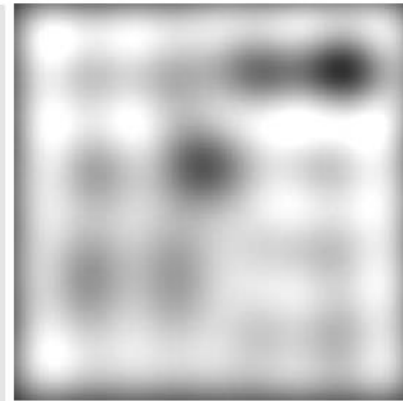
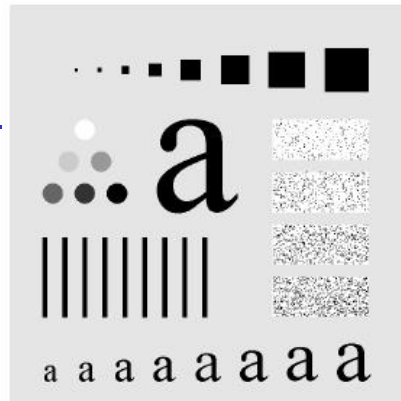


a b

FIGURE 4.41 (a) Test pattern of size 688×688 pixels, and (b) its Fourier spectrum. The spectrum is double the image size due to padding but is shown in half size so that it fits in the page. The superimposed circles have radii equal to 10, 30, 60, 160, and 460 with respect to the full-size spectrum image. These radii enclose 87.0, 93.1, 95.7, 97.8, and 99.2% of the padded image power, respectively.

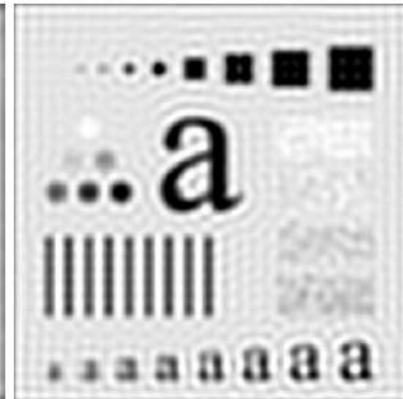
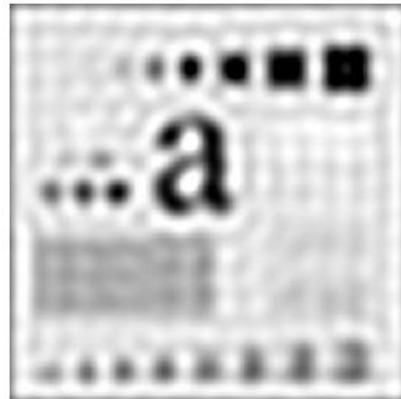
Applying the ILPF – Blurring and Ringing

Original



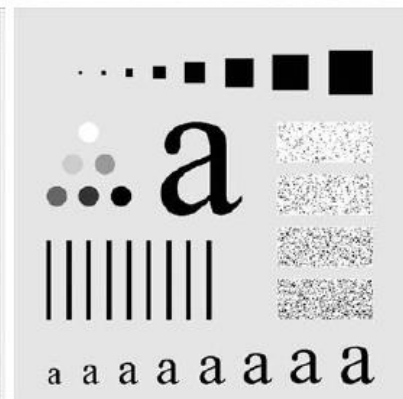
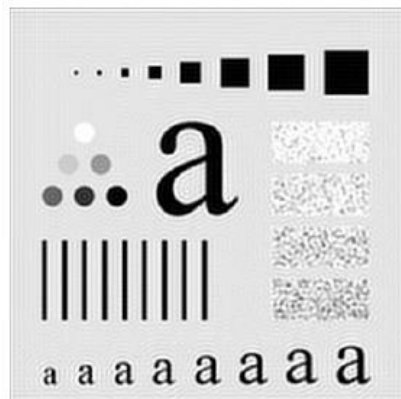
ILPF, cutoff 10,
Energy 87%

ILPF, cutoff 30
Energy 93.1%



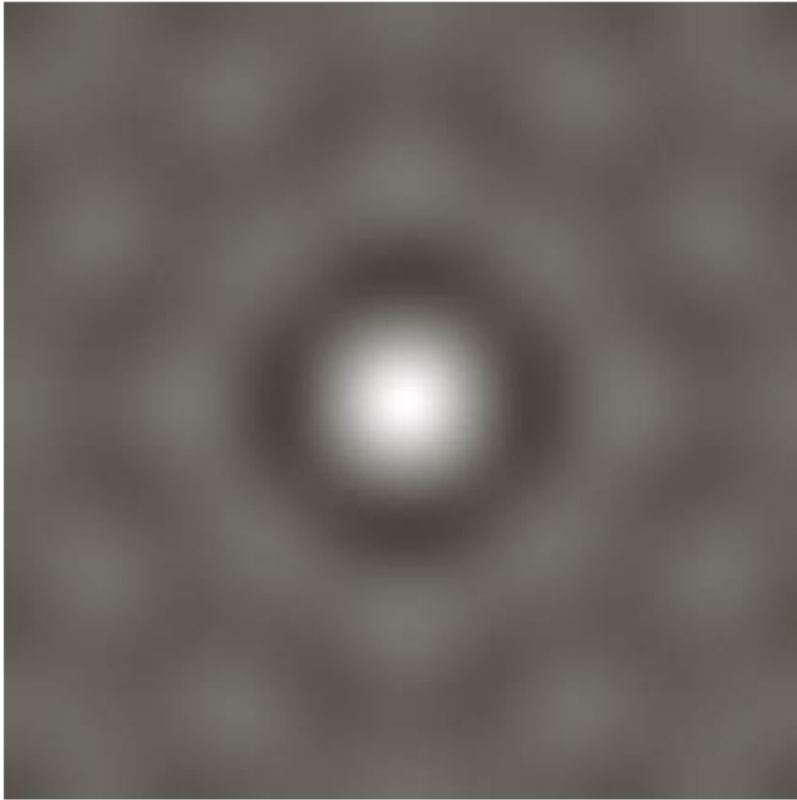
ILPF, cutoff 60
Energy 95.7%

ILPF, cutoff 160
Energy 97.8%



ILPF, cutoff 460
Energy 99.2%

Why?



a b

FIGURE 4.43

(a) Representation in the spatial domain of an ILPF of radius 5 and size 1000×1000 .

(b) Intensity profile of a horizontal line passing through the center of the image.

Butterworth Lowpass Filters (BLPF)

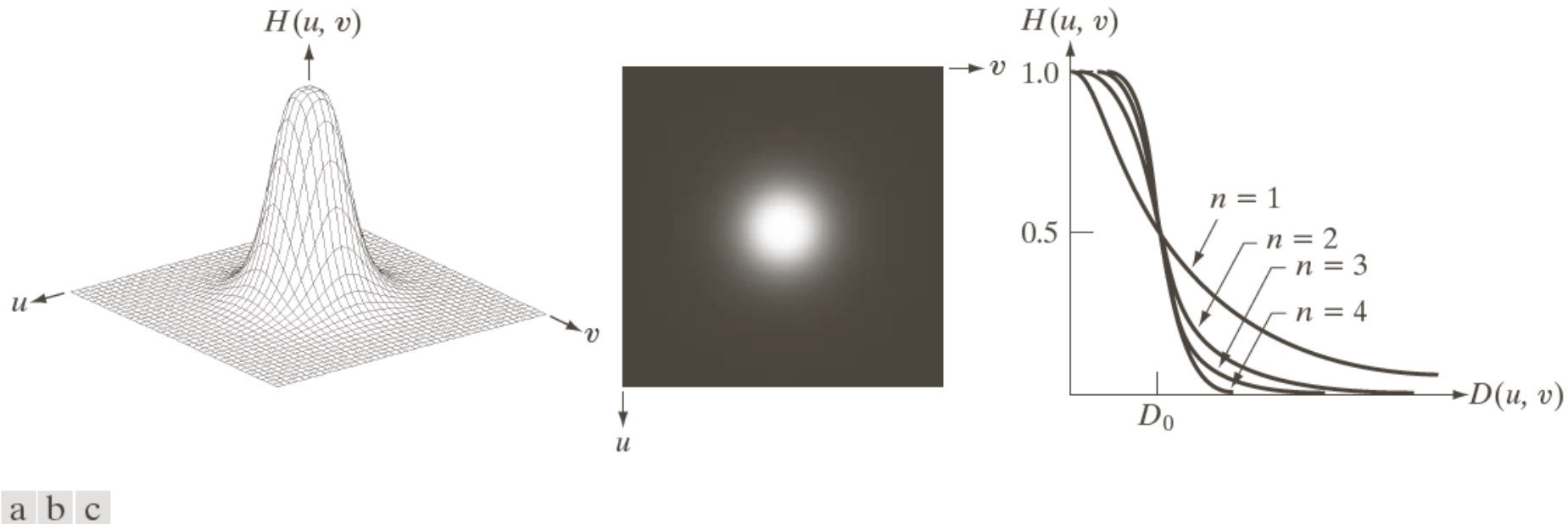


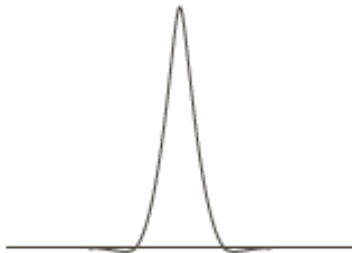
FIGURE 4.44 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

$$H(u, v) = \frac{1}{1 + [D(u, v) / D_0]^{2n}} \rightarrow \text{order}$$

D_0 is the cutoff frequency

Applying the BLPF

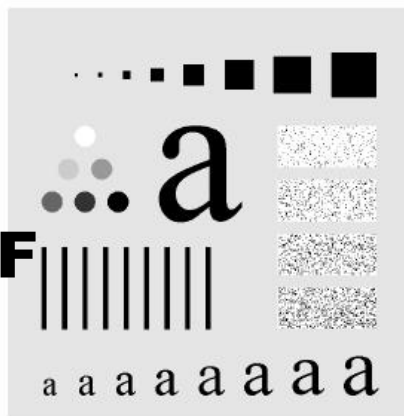
Original



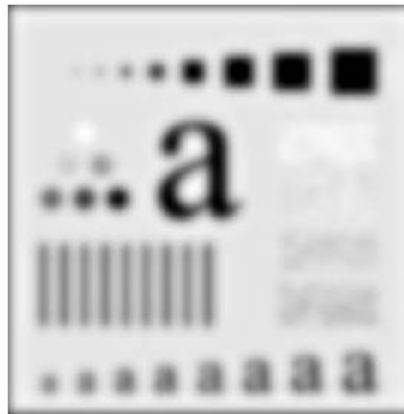
Order 2

ILPF, cutoff 30
Energy 93.1%

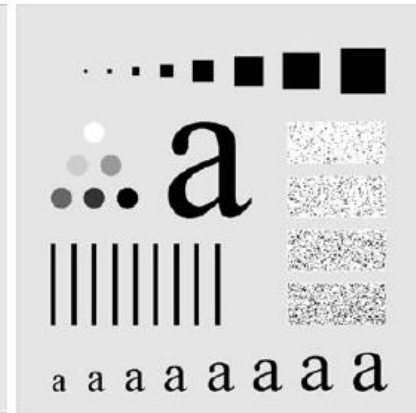
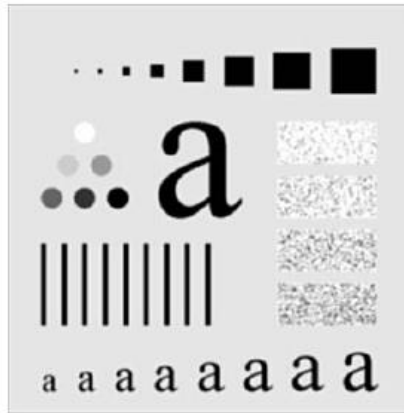
ILPF, cutoff 160
Energy 97.8%



ILPF, cutoff 10
Energy 87%



ILPF, cutoff 60
Energy 95.7%



ILPF, cutoff 460
Energy 99.2%

a b
c d
e f

FIGURE 4.45 (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.

Different-Order BLPF

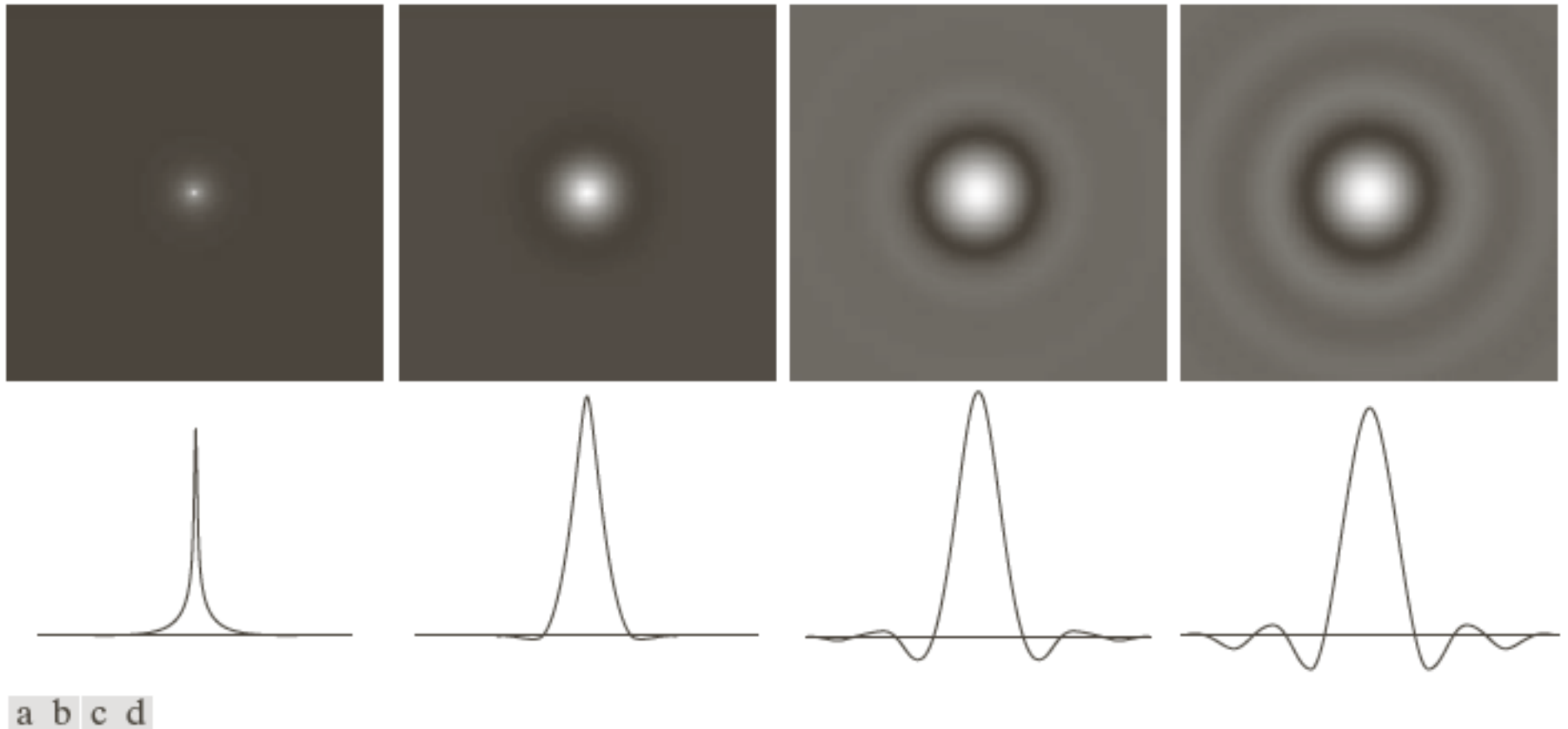
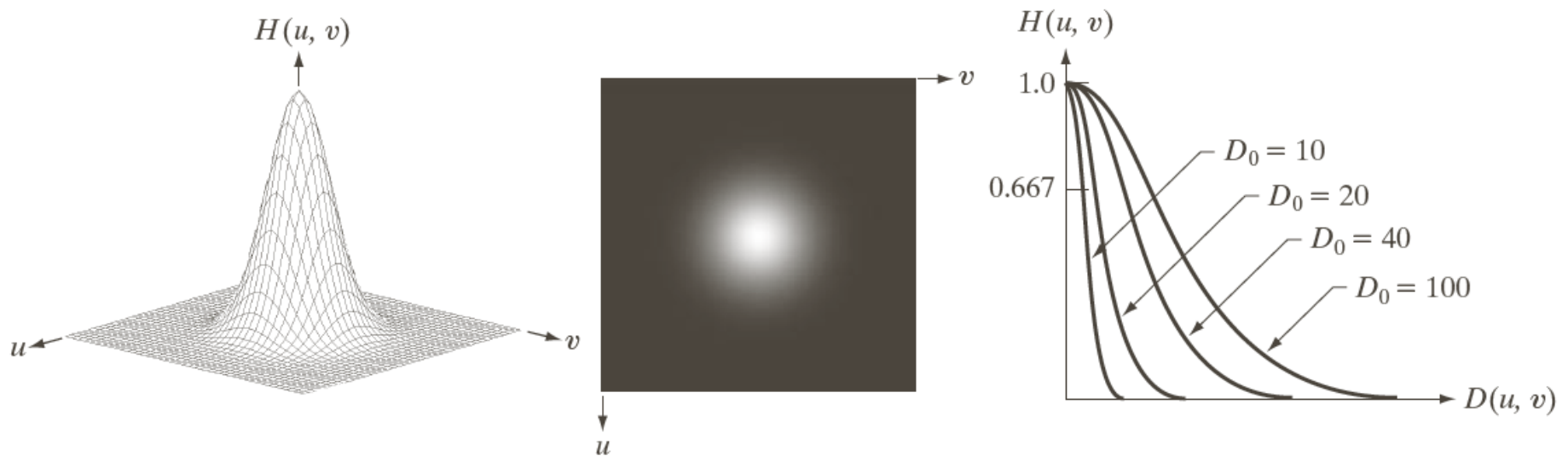


FIGURE 4.46 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding intensity profiles through the center of the filters (the size in all cases is 1000×1000 and the cutoff frequency is 5). Observe how ringing increases as a function of filter order.

Gaussian Lowpass Filters



a b c

FIGURE 4.47 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0 .

$$H(u, v) = e^{-D^2(u, v) / 2D_0^2}$$

Applying the GLPF

D0 = cutoff radius



a b
c d
e f

BLPF with order 2

FIGURE 4.45 (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.



a b
c d
e f

GLPF

FIGURE 4.48 (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs. 4.42 and 4.45.

Summary of Lowpass Filters

TABLE 4.4

Lowpass filters. D_0 is the cutoff frequency and n is the order of the Butterworth filter.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$	$H(u, v) = \frac{1}{1 + [D(u, v)/D_0]^{2n}}$	$H(u, v) = e^{-D^2(u,v)/2D_0^2}$

Applications

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



ea

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



ea

a b

FIGURE 4.49

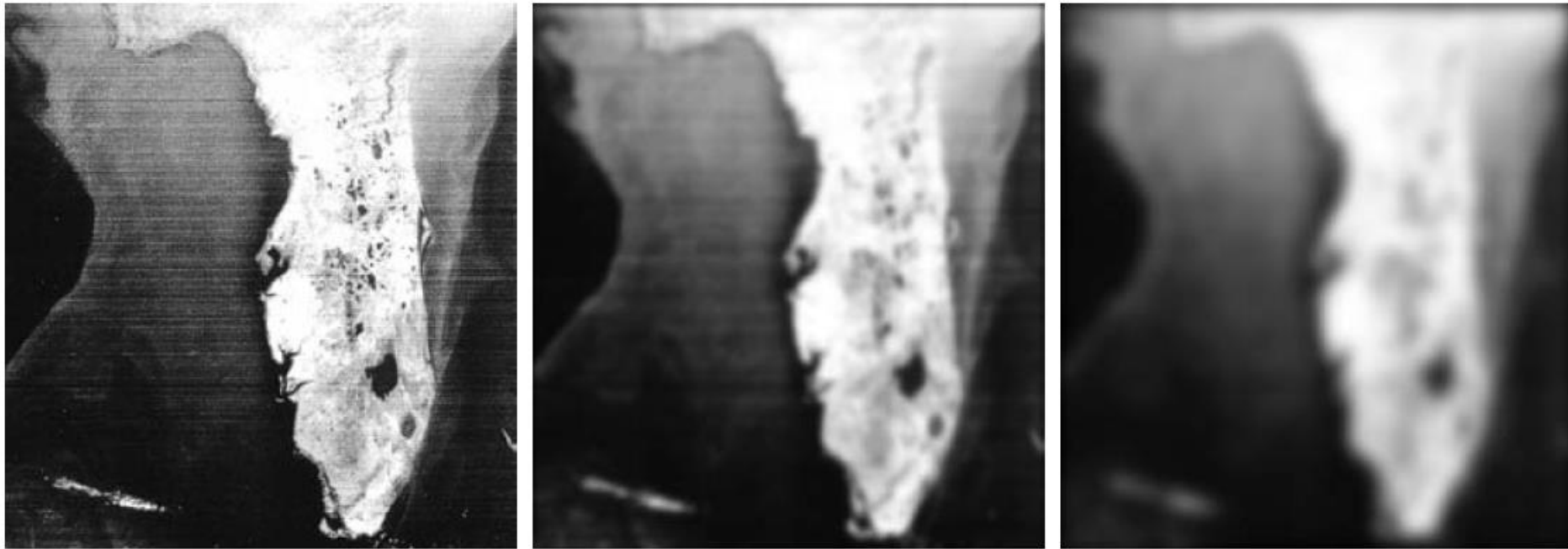
(a) Sample text of low resolution (note broken characters in magnified view).
(b) Result of filtering with a GLPF (broken character segments were joined).

Applications



FIGURE 4.50 (a) Original image (784×732 pixels). (b) Result of filtering using a GLPF with $D_0 = 100$. (c) Result of filtering using a GLPF with $D_0 = 80$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

Applications



a b c

FIGURE 4.51 (a) Image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

Highpass Filters in Frequency Domain

$$H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

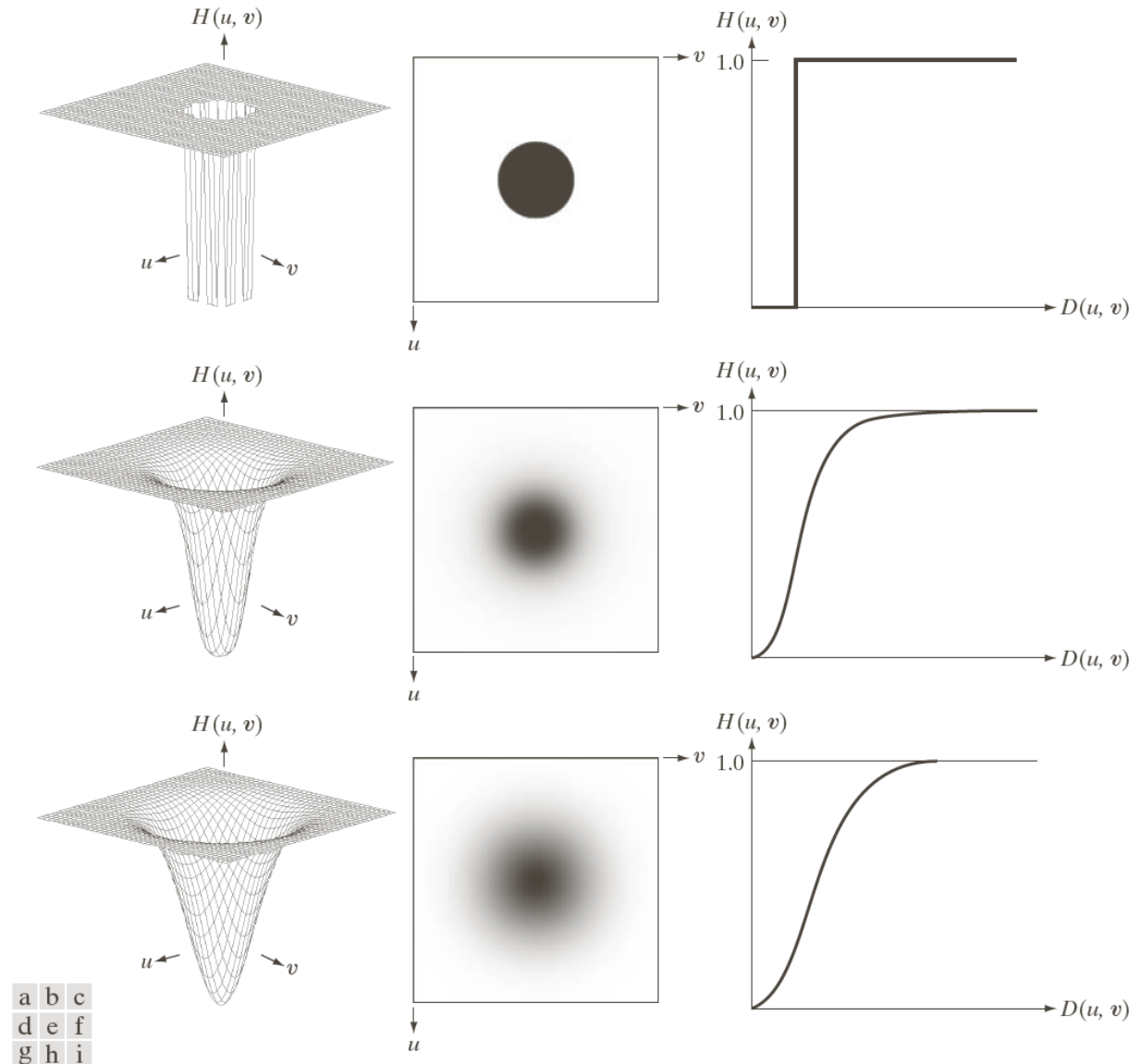


FIGURE 4.52 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

Highpass Filters

TABLE 4.5

Highpass filters. D_0 is the cutoff frequency and n is the order of the Butterworth filter.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$	$H(u, v) = \frac{1}{1 + [D_0/D(u, v)]^{2n}}$	$H(u, v) = 1 - e^{-D^2(u,v)/2D_0^2}$

Highpass Filters In Spatial Domain

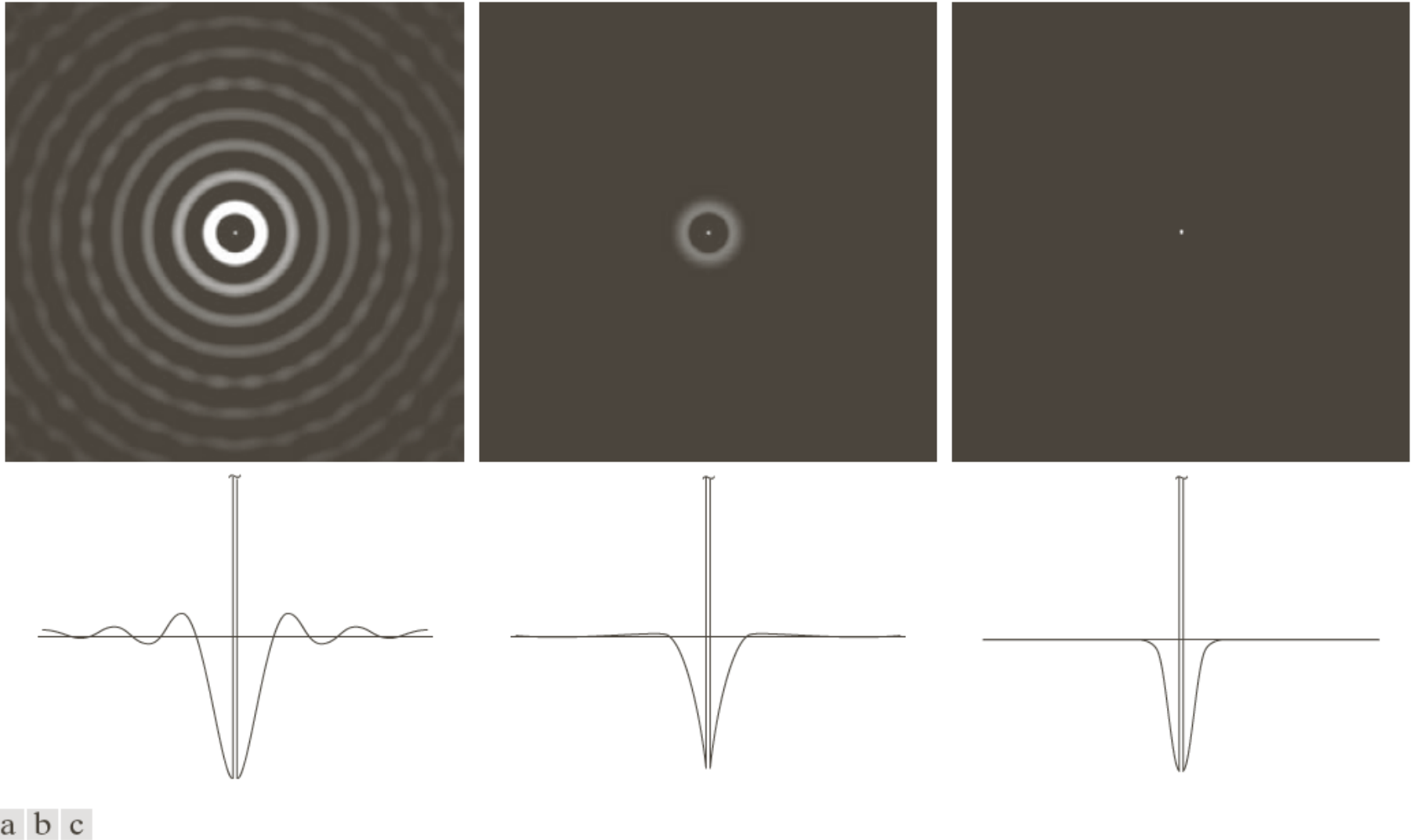


FIGURE 4.53 Spatial representation of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding intensity profiles through their centers.

Applying IHPF



a b c

FIGURE 4.54 Results of highpass filtering the image in Fig. 4.41(a) using an IHPF with $D_0 = 30, 60$, and 160 .

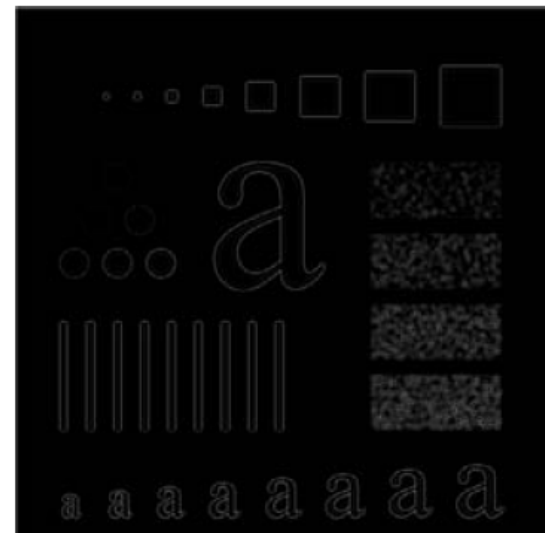
Applying BHPF/GHPF

$D_0 = 30$

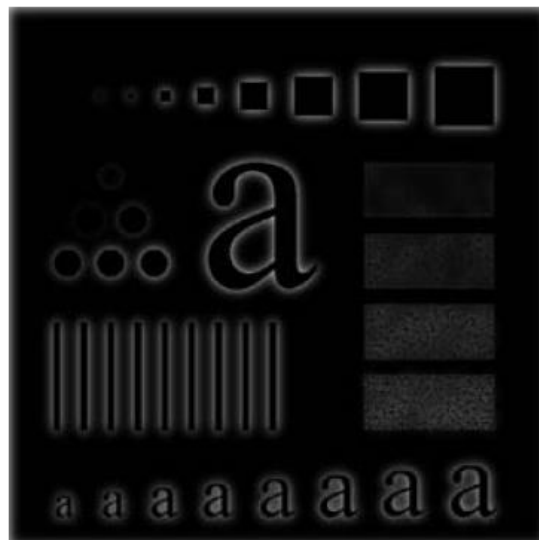
$D_0 = 60$

$D_0 = 160$

BHPF



GHPF



Applications



a b c

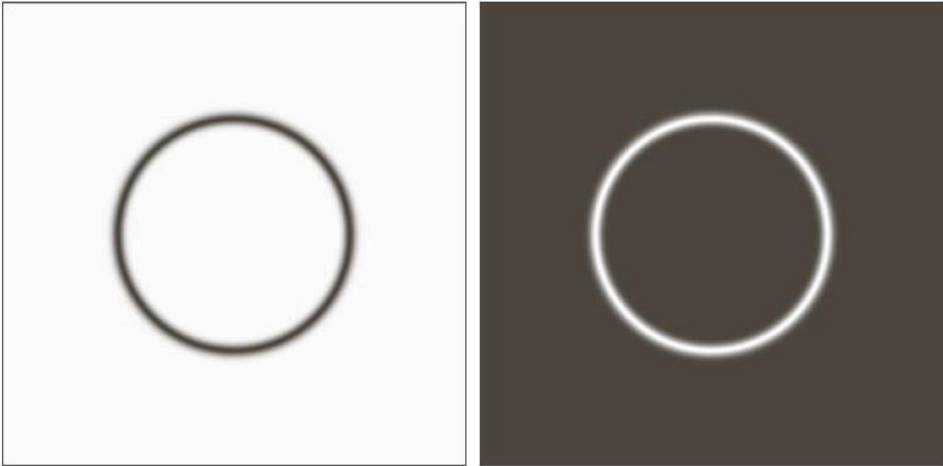
FIGURE 4.57 (a) Thumb print. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

Bandreject and Bandpass Filters

TABLE 4.6

Bandreject filters. W is the width of the band, D is the distance $D(u, v)$ from the center of the filter, D_0 is the cutoff frequency, and n is the order of the Butterworth filter. We show D instead of $D(u, v)$ to simplify the notation in the table.

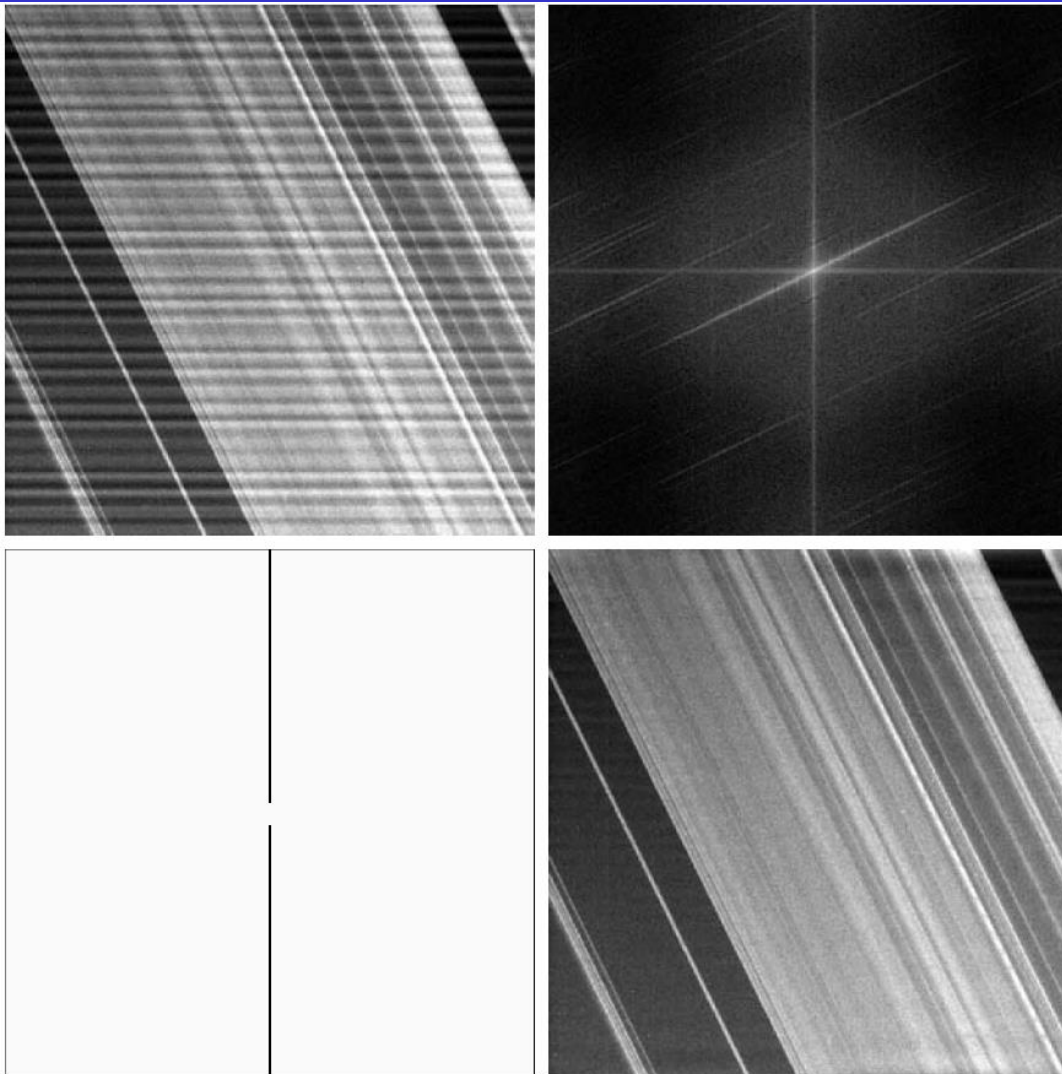
Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u, v) = \frac{1}{1 + \left[\frac{DW}{D^2 - D_0^2} \right]^{2n}}$	$H(u, v) = 1 - e^{-\left[\frac{D^2 - D_0^2}{DW} \right]^2}$



a b

FIGURE 4.63
 (a) Bandreject Gaussian filter.
 (b) Corresponding bandpass filter.
 The thin black border in (a) was added for clarity; it is not part of the data.

Example



a	b
c	d

FIGURE 4.65

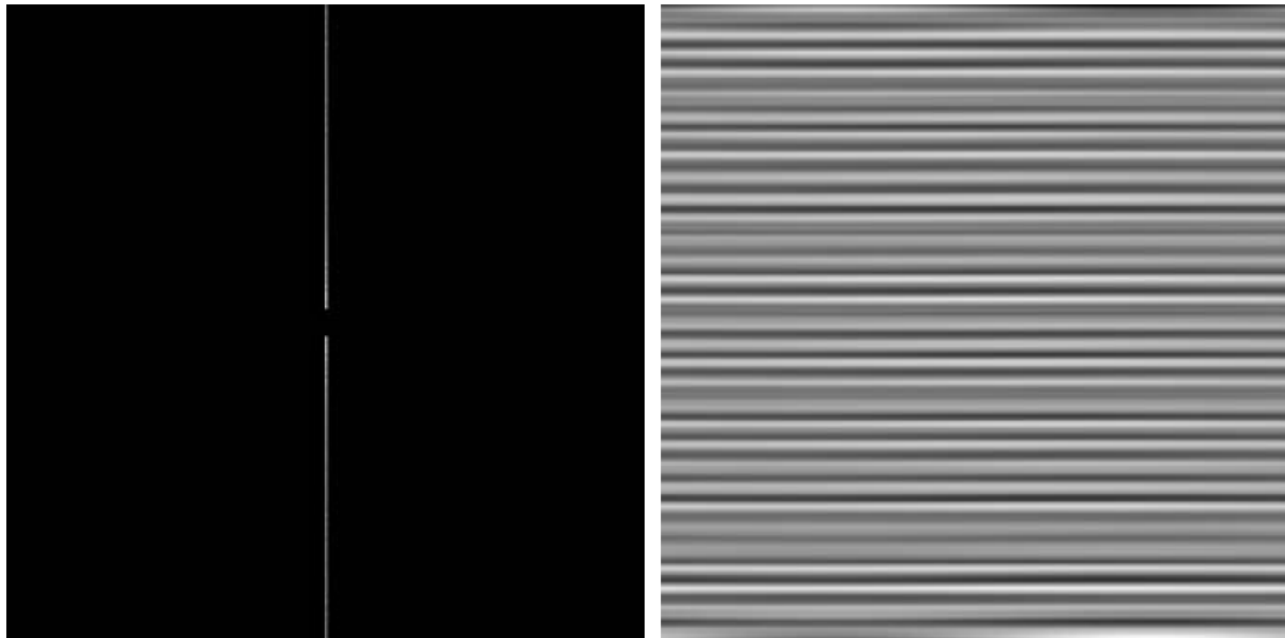
(a) 674×674 image of the Saturn rings showing nearly periodic interference.

(b) Spectrum: The bursts of energy in the vertical axis near the origin correspond to the interference pattern.

(c) A vertical notch reject filter.

(d) Result of filtering. The thin black border in (c) was added for clarity; it is not part of the data. (Original image courtesy of Dr. Robert A. West, NASA/JPL.)

Example



a b

FIGURE 4.66

(a) Result (spectrum) of applying a notch pass filter to the DFT of Fig. 4.65(a).
(b) Spatial pattern obtained by computing the IDFT of (a).

Implementation

Separability of the 2D DFT

Computing the IDFT using a DFT algorithm

The Fast Fourier Transform (read Chapter 4.11.3)

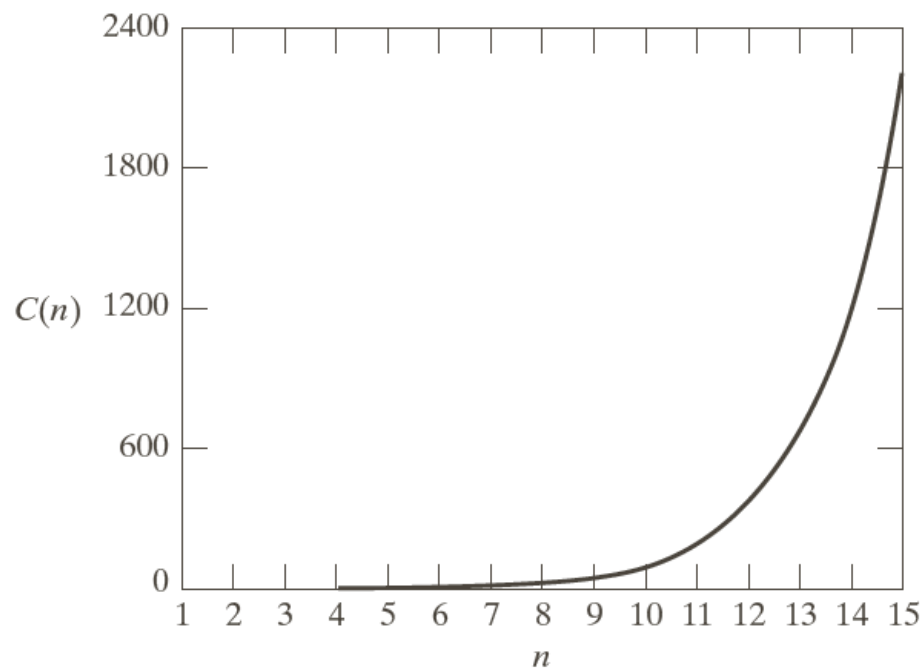


FIGURE 4.67
Computational advantage of the FFT over a direct implementation of the 1-D DFT. Note that the advantage increases rapidly as a function of n .

Reading Assignments

Chapter 4.3 – 4.11

Image Degradation and Restoration



http://fireoracleproductions.com/services__samp

Image degradation due to

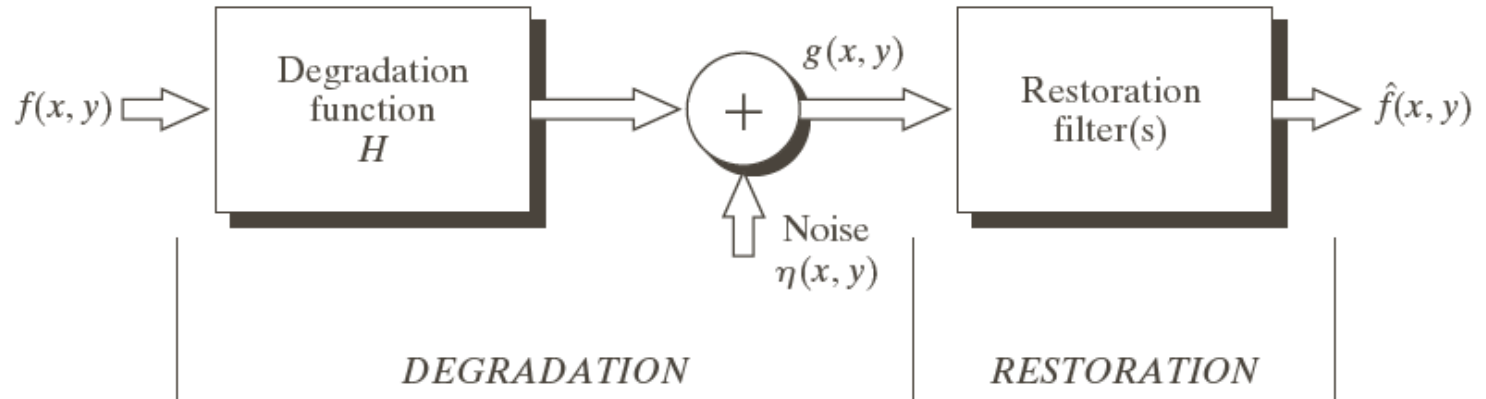
- noise in transmission
- imperfect image acquisition
 - environmental condition
 - quality of sensor



Google image

Image Degradation and Restoration

FIGURE 5.1
A model of the
image
degradation/
restoration
process.



$$g(x, y) = h(x, y) \otimes f(x, y) + \eta(x, y)$$

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

Identity $H(u, v) \rightarrow$ degradation only comes from additive noise

Restoration in the Presence of Noise Only – Spatial Filtering

$$g(x, y) = f(x, y) + \eta(x, y)$$

$$G(u, v) = F(u, v) + N(u, v)$$

Image Restoration with Additive Noise

$$g(x, y) = f(x, y) + \eta(x, y)$$

$$G(u, v) = F(u, v) + N(u, v)$$

Noise models:

- Impulse noise: pepper and salt
- Continuous noise model:
 - Gaussian, Rayleigh, Gamma, Exponential, Uniform

Properties of Noise

- **Spatial properties**
 - Spatially periodic noise
 - Spatially independent noise
- **Frequency properties**
 - White noise – noise containing all frequencies within a bandwidth